# Monetary Economics and Policy

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## Chapter 10: Deleveraging and Credit Crunch

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# What We Will Learn

- **Endogenous origins of liquidity traps:** How deleveraging and credit crunches can drive the natural real rate down (and potentially below zero), even without exogenous patience shocks.
- **Deleveraging mechanism:** Why forced repayment cuts borrowers' spending and requires savers to spend more for goods markets to clear—lowering the natural rate.
- **Credit-crunch mechanism:** How banking stress tightens lending, widens credit spreads, and depresses demand—again pushing the natural rate down.
- How debt and spreads reshape the **aggregate demand (IS/AD) relation**, including Fisher debt-deflation versus borrowing-capacity effects.
- Why **sticky prices** and the **zero lower bound** can prevent the policy rate from falling enough, producing a Keynesian slump.
- What this implies for policy: why central banks must **monitor spreads** and may need **forward guidance** and **credit easing** to shorten liquidity traps.

# Outline

# Liquidity Trap: Two Financial Mechanisms

- Chapter 9 emphasized **exogenous preference shifts** (higher patience) as a driver of a low natural real rate, potentially pushing the economy into a liquidity trap.
- The post-2008 literature highlights additional **financial crisis mechanisms** that can also drive the natural real rate negative:
  1. **Household debt deleveraging**.
  2. **Banking sector turbulence / credit crunch**.

# Debt Deleveraging

- Deleveraging typically follows a leveraging boom:
  - Excessive optimism leads households to borrow and spend aggressively.
- At the **"Minsky moment"** (Eggertsson and Krugman 2012), agents recognize that debt is unsustainable:
  1. New borrowing stops.
  2. Overextended borrowers must repay (cut consumption).
- The demand for Full-employment then requires savers/creditors to spend more as debtors cut spending $\Rightarrow$ the **natural real rate falls**.
- If deleveraging is rapid, the natural real rate can become **negative**. With a limited ability to cut nominal rates, the **ZLB binds** and the output falls below potential.

# Banking Turbulence and Credit Spreads

- Interbank stress raises banks' funding costs, often reflecting:
  - Capital losses.
  - Leverage reductions / tighter regulatory or market constraints.
- Tighter constraints make banks less willing to lend $\Rightarrow$ **credit crunch** and downturn.
- Banking turbulence (and deleveraging) typically widens **credit spreads**:
  - Borrowing becomes more expensive than saving/lending.
  - Private demand weakens.
- To sustain demand at full employment, the **natural (risk-free) real rate must fall** to offset higher borrowing costs.
- Stabilization then requires a lower policy rate; at sufficiently large spreads, the economy risks hitting the **zero lower bound**.

# Policy Implications of Credit Spreads

- If the natural rate is shifted by credit spreads, policy must account for **credit-market conditions** (not only inflation and output gaps).
- Credit spreads are **endogenous**:
  - Depend on macro conditions and balance-sheet strength (bank equity, borrower repayment capacity).
  - Can amplify shocks via tighter lending conditions.
- Monetary policy affects spreads through:
  1. Conventional policy-rate cuts.
  2. At the ZLB, **unconventional credit easing** that improves balance sheets and compresses spreads.
- By limiting spread increases, policy prevents the natural rate from falling as much, reducing the likelihood and severity of a liquidity trap.

# Outline of the Results

- The chapter develops a heterogeneous-agent New Keynesian model with two types of agents: *savers* and *borrowers*.
- A forced reduction in debt/loans (deleveraging or a credit contraction) can drive the **natural real rate** into negative territory.
- More generally, movements in the natural rate are driven by **credit-market spreads**, which in turn depend on macro-financial conditions and policy.
- Monetary policy can shorten liquidity traps, including through **unconventional credit easing**.
- Optimal policy is welfare-based and internalizes **distributional consequences** across agents.

# Preferences

- The economy is inhabited by savers $s$ and borrowers $b$, each receiving a constant endowment $\frac{1}{2}Y$ and with savers being more patient than borrowers $\beta^s > \beta^b$.

**Expected Intertemporal Utility**

$$\mathbb{E}_{t_0} \sum_{t=t_0}^{\infty} (\beta^j)^{t-t_0} \log C_t^j, \qquad j \in \{s, b\}. \tag{10.1}$$

**Flow Budget Constraint**

$$B_t^j = (1 + r_{t-1}) B_{t-1}^j - C_t^j + \frac{1}{2} Y, \qquad j \in \{s, b\} \tag{10.2}$$

- $B_t^j$ are real assets and $r_{t-1}$ is the real interest rate set in period $t-1$.

# Borrowing Limit and Euler Equation

**Borrowing Limit**

$$-(1+r_{t-1})B_{t-1}^{j} \leq B^{high} \leq \frac{1}{2}\frac{\beta}{1-\beta}Y. \tag{10.3}$$

- Outstanding debt and interest payment cannot exceed $B^{high}$, while $\frac{1}{2}\frac{\beta}{1-\beta}Y$ represents the maximum amount of debt that an agent can repay with certainty (see Chapter 1).
- Note: We are setting $\beta^{s} = \beta$.
- Optimal consumption implies the Euler equation:

$$\frac{1}{C_t^{j}} \geq \beta^{j}(1+r_t)\mathbb{E}_t\left\{\frac{1}{C_{t+1}^{j}}\right\}, \tag{10.4}$$

which holds with inequality when the borrowing limit is binding at $t$.

# Steady State

- Borrowers are less patient and therefore hit their borrowing limit. Equation (10.2) together with the borrowing limit then implies:

$$C^b = \frac{1}{2}Y - \frac{r}{1+r}B^{high}.$$

- Savers are unconstrained and therefore their Euler equation (10.4) holds with equality. In steady state: $1 + r = \frac{1}{\beta^s} = \frac{1}{\beta}$. Moreover, since $C^b + C^s = Y$:

$$C^b = \frac{1}{2}Y + \frac{r}{1+r}B^{high}.$$

# Deleveraging Shock

- Consider a shock at time $t$ that forces debtors to repay their debt so that $B^{high}$ goes to a lower threshold $B^{low}$ with $B^{high} > B^{low}$.
- At $t+1$ the economy reaches the new steady state (note that $\frac{r}{1+r} = 1-\beta$):

$$C^b_{t+1} = \frac{1}{2}Y - (1-\beta)B^{low}, \quad C^s_{t+1} = \frac{1}{2}Y + (1-\beta)B^{low}.$$

- Using the flow budget constraint (10.2) and goods market equilibrium $C^b + C^s = Y$ we can obtain consumption at time $t$:

$$C^b_t = \frac{1}{2}Y + \frac{B^{low}}{1+r_t} - B^{high}, \quad C^s_t = \frac{1}{2}Y + B^{high} - \frac{B^{low}}{1+r_t}.$$

# The Short-Run Real Rate

- Using the savers' Euler equation (10.4) between $t$ and $t+1$, the market-clearing (natural) short-run real rate is:

$$1 + r_t = \frac{1}{\beta} \frac{\frac{1}{2}Y + B^{low}}{\frac{1}{2}Y + B^{high}} < \frac{1}{\beta}.$$

- Hence $r_t$ falls relative to its initial value and, for a sufficiently large deleveraging shock,

$$\beta B^{high} - B^{low} > \frac{Y(1-\beta)}{2\beta},$$

  the natural real rate becomes negative.

- **Intuition:** Deleveraging forces borrowers to cut consumption (increase saving). To keep output at full capacity, savers must be induced to absorb the excess supply of goods by consuming more, which requires a *lower* real interest rate.

# Zero Lower Bound

- The $r_t$ derived above is the **frictionless market-clearing** (natural) real rate.
- With nominal rigidities, the actual real rate may not fall enough to match the natural rate.
- If the natural real rate turns negative, a nominal policy rate constrained by the ZLB, $i_t \geq 0$, can be too high in real terms:

$$r_t \;\approx\; i_t - \mathbb{E}_t \pi_{t+1}.$$

- **Consequence:** Insufficiently low real rates with respect to the natural rate of interest depress aggregate demand, producing a Keynesian slump (output falls below potential).

# Preferences

- The economy is populated by two types of agents:
  - **Savers** $s$: mass $\chi$,
  - **Borrowers** $b$: mass $1-\chi$.

**Intertemporal Utility**

$$\mathbb{E}_{t_0}\sum_{t=t_0}^{\infty}\beta^{t-t_0}\Big(U(C_t^j)-H(L_t^j)\Big),\qquad j\in\{s,b\}. \tag{10.5}$$

- $U(\cdot)$ is concave in consumption $C$,
- $H(\cdot)$ is convex in labor $L$.
- We adopt exponential utility in consumption:

$$U(C_t^j)=1-\exp(-z\,C_t^j),\qquad z>0.$$

# Flow Budget Constraint

**Flow Budget Constraint**

$$B_t^j = (1 + i_{t-1})B_{t-1}^j - P_t C_t^j + W_t^j L_t^j + \Psi_t - T_t^j. \tag{10.6}$$

**Interpretation:**

- $B_t^j$ is the nominal position in one-period risk-free bonds.
- $i_{t-1}$ is the nominal interest rate.
- $P_t$ is the price level.
- $W_t^j$ and $L_t^j$ are the wage and labor of type $j$.
- $\Psi_t$ denotes firms' profits (distributed equally across agents).
- $T_t^j$ are lump-sum taxes.

# Safe-Debt Threshold

- As before, there is a threshold below which debt can be treated as **safe**:

$$-(1 + r_t)\frac{B_t^j}{P_t} \leq \bar{B}, \tag{10.7}$$

  where $r_t$ is the real interest rate.

- We assume that the safe-debt limit tightens at time $t$:

$$\bar{B}: \; B^{high} \longrightarrow B^{low}.$$

# Linearized Euler Equation

- Households choose consumption and labor to maximize utility (10.5) subject to the flow budget constraint (10.6), taking into account the debt limit (10.7).
- For savers, under perfect foresight, the Euler equation is

$$\exp(-zC_t^s) = \beta(1+i_t)\frac{P_t}{P_{t+1}}\exp(-zC_{t+1}^s).$$

- Taking logs and rearranging yields

$$C_t^s = C_{t+1}^s - \tilde{\sigma}\Big[\ln(1+i_t) - \pi_{t+1} - \tilde{\beta}\Big], \tag{10.8}$$

  where $\pi_{t+1} \equiv \ln(P_{t+1}/P_t)$, $\tilde{\beta} \equiv -\ln\beta$, and $\tilde{\sigma} \equiv 1/z$.
- **Remark:** With exponential utility, consumption enters in *levels* (not logs) in (10.8).

# AD Equation (Aggregation)

- Using the aggregate resource constraint

$$Y_t = \chi C_t^s + (1-\chi) C_t^b,$$

substitute (10.8) to obtain the aggregate demand (AD) relation:

$$Y_t = Y_{t+1} - \chi\tilde{\sigma}\Big[\ln(1+i_t) - \pi_{t+1} - \tilde{\beta}\Big] + (1-\chi)\big(C_t^b - C_{t+1}^b\big).$$

- Expressing the equation in log deviations from steady state gives:

$$\hat{Y}_t = \hat{Y}_{t+1} - \chi\sigma\Big[\hat{\imath}_t - (\pi_{t+1} - \pi)\Big] + (1-\chi)\left(\frac{C_t^b - C_{t+1}^b}{Y}\right), \tag{10.9}$$

where $\hat{Y}_t \equiv \ln(Y_t/Y)$ and

$$\hat{\imath}_t \equiv \ln\left(\tfrac{1+i_t}{1+i}\right), \qquad \ln(1+i) = \tilde{\beta} + \pi, \qquad \sigma \equiv \frac{\tilde{\sigma}}{Y} = \frac{1}{zY}.$$

# AD Equation (Borrower Deleveraging Term)

- The borrower term $(C_t^b - C_{t+1}^b)$ shifts aggregate demand:
  - Deleveraging lowers borrowers' short-run consumption $\Rightarrow$ AD falls even if interest rates are unchanged.
- **Assumptions:** Borrowers are always at the debt limit $\bar{B}: \ B^{high} \longrightarrow B^{low}$ at $t$, the real rate equals $r$ from $t+1$ onward, and $W_t^j L_t^j + \Psi_t = P_t Y_t$ for all $t$.
- Borrower consumption (from 10.6):

$$C_t^b = -\frac{1+i_{t-1}}{1+r_{t-1}} \frac{P_t}{P_{t-1}} B^{high} + \frac{B^{low}}{1+r_t} + Y_t - T_t^b,$$

$$C_{t+1}^b = -\frac{1+i_t}{1+r_t} \frac{P_{t+1}}{P_t} B^{low} + \frac{B^{low}}{1+r} + Y_{t+1} - T_{t+1}^b.$$

# AD Equation

- Approximating the borrower-consumption expressions around the initial debt position and substituting into (10.9) yields the short-run AD equation:

$$\hat{Y}_t = \mathbb{E}_t \hat{Y}_{t+1} - \varphi_r \left[ \hat{\imath}_t - (p_{t+1} - p_t - \pi) \right] + \frac{1-\chi}{\chi} \left[ \hat{b}_t + b(p_t - p^*) \right] - \frac{1-\chi}{\chi} \left( \hat{T}_t^b - \mathbb{E}_t \hat{T}_{t+1}^b \right), \tag{10.10}$$

where

$$\varphi_r \equiv \frac{\sigma\chi + (1-\chi) b\beta}{\chi} \geq 0, \qquad b \equiv \frac{B^{high}}{Y}, \qquad \hat{b}_t \equiv \frac{B^{low} - B^{high}}{Y}, \qquad \hat{T}_t^b \equiv \frac{T_t^b - T^b}{Y}.$$

- Here $Y$ is steady-state output, $p_t$ is the log price level, and inflation is $\pi_t \equiv p_t - p_{t-1}$ (with target $\pi$).

# How Debt Changes the Slope of AD

- Relative to the benchmark AD curve (Chapter 6), debt introduces **two opposing forces** that affect the slope of AD in $(p_t, y_t)$ space:
  1. **Real-rate / borrowing-capacity channel:** higher $p_t$ raises the real rate, tightens the borrowing constraint, reduces $C_t^b$, and lowers demand.
     - This increases $\varphi_r$ and tends to **flatten** AD.
  2. **Fisher debt-deflation channel** (Fisher 1933): higher $p_t$ reduces the real value of nominal debt, raises borrowers' net worth, increases $C_t^b$, and raises demand.
     - This adds the term $\frac{1-\chi}{\chi} b\,(p_t - p^*)$ and tends to **steepen** AD.
- **Net effect:** If the Fisher channel dominates, AD becomes **steeper** than in the previous model.

# Labor Supply: Intratemporal Condition

- For each type $j \in \{s, b\}$, labor supply satisfies the intratemporal condition (MRS between labor and consumption equals the real wage):

$$\frac{H_l(L_t^j)}{U_c(C_t^j)} = \frac{W_t^j}{P_t}. \tag{10.11}$$

- **Functional forms.** Assume isoelastic disutility of work and exponential utility of consumption:

$$H(L^j) = \frac{(L^j)^{1+\eta}}{1+\eta} \Rightarrow H_l(L_t^j) = (L_t^j)^\eta, \qquad U(C^j) = 1 - e^{-zC^j} \Rightarrow U_c(C_t^j) = ze^{-zC_t^j}.$$

- Hence, for each $j$,

$$\frac{(L_t^j)^\eta}{z\, e^{-zC_t^j}} = \frac{W_t^j}{P_t}.$$

# Labor Supply: Aggregation

- **Aggregation.** Output is produced with

$$Y_t = A_t L_t, \qquad L_t = (L_t^s)^{\chi}(L_t^b)^{1-\chi},$$

  and the corresponding aggregate wage index is

$$W_t = (W_t^s)^{\chi}(W_t^b)^{1-\chi}.$$

- Taking a weighted average of the two intratemporal conditions (for $j = s, b$) with weights $\chi$ and $1-\chi$ gives the aggregate labor-supply schedule:

$$\frac{L_t^{\eta}}{z\exp\big[-z(\chi C_t^s + (1-\chi)C_t^b)\big]} = \frac{W_t}{P_t}. \tag{10.12}$$

- Using goods-market clearing $Y_t = \chi C_t^s + (1-\chi)C_t^b$ and production $Y_t = A_t L_t$, (10.12) simplifies to:

$$\frac{(Y_t/A_t)^{\eta}}{z\exp(-zY_t)} = \frac{W_t}{P_t}. \tag{10.13}$$

# Natural Output

- Under flexible prices, firms set a markup over marginal cost:

$$P_t = \mu_t \frac{W_t}{A_t}.$$

- Combining this with (10.13), the natural level of output $Y_t^n$ is implicitly defined by:

$$\frac{\left(\frac{Y_t^n}{A_t}\right)^\eta}{z\exp(-zY_t^n)} = \frac{A_t}{\mu_t}.$$

- **Log-linear approximation (Chapter 5, with $G_t = 0$):**

$$\hat{Y}_t^n = \frac{1+\eta}{\sigma^{-1}+\eta}\hat{A}_t - \frac{1}{\sigma^{-1}+\eta}\hat{\mu}_t.$$

# AS & AD Equations

- Given the natural output definition, the AS block is the same as in Chapter 5 (NKPC):

$$\pi_t - \pi = \kappa(\hat{Y}_t - \hat{Y}_t^n) + \beta E_t(\pi_{t+1} - \pi),$$

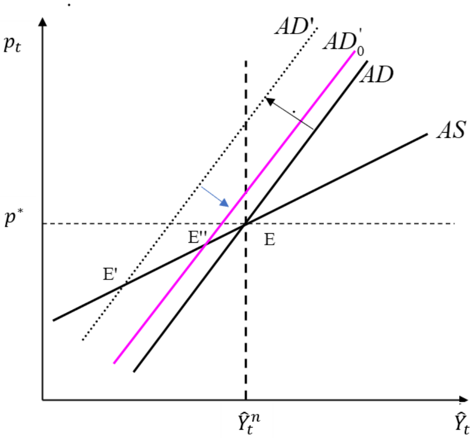
with

$$\kappa \equiv \frac{1-\alpha}{\alpha}(1-\alpha\beta)\,\frac{\sigma^{-1}+\eta}{1+\theta\eta}.$$

- For tractability, assume $\pi_T = \pi$ for all $T \geq t+1$.
  - Then $\hat{Y}_T = \hat{Y}_T^n$ for all $T \geq t+1$.
- Under these assumptions, the time-$t$ AD schedule is upward sloping and can be written as:

$$\hat{Y}_t = \hat{Y}_{t+1}^n - \varphi_r\,\hat{r}_t - \frac{1-\chi}{\chi}\left(\hat{\tau}_t^b - \hat{T}_{t+1}^b\right) + \frac{1-\chi}{\chi}\left[\hat{b}_t + b\left(p_t - p^*\right)\right].$$

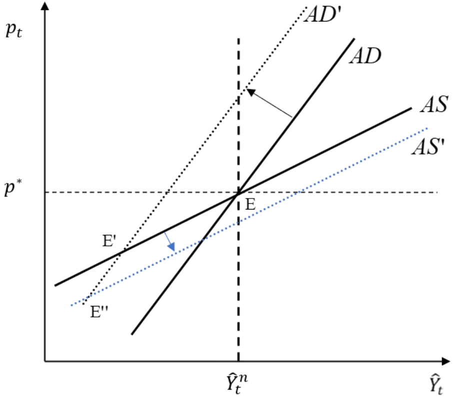
# Deleveraging Shock




Deleveraging shifts $AD$ left ($E \rightarrow E'$). Cutting $i_t$ can move the economy to $E''$, but may be limited by the ZLB.

- Deleveraging is a **left shift** of $AD$ (in (10.10): $\hat{b}_t < 0$): $E \longrightarrow E'$, with **lower output and prices**.
- Lowering the policy rate helps, but at the ZLB conventional policy may only move the economy to $E''$ (still below the initial allocation).
- **Paradox of thrift:** when borrowers try to save more, AD falls and output contracts, so *aggregate* saving can fall. (Eggertsson and Krugman 2012)

# Paradox of "Toil"




$AD$ shifts up ($E \to E'$). In a liquidity trap, a downward shift in $AS$ can be contractionary ($E' \to E''$).

- In the liquidity-trap region, an **upward** shift in $AD$ moves equilibrium $E \to E'$.
- A **downward** shift in $AS$ can *aggravate* the recession, moving equilibrium $E' \to E''$.
- This reversal is the **paradox of "toil"**: changes that would normally raise output can become contractionary when monetary policy is constrained.
  - Examples: temporary increase in productivity, or a fall in the markup.

# Paradox of Flexibility and Escaping the Liquidity Trap

**Paradox of flexibility**

- More flexible prices $\Rightarrow$ a **steeper** AS curve.
- With deleveraging, a steeper AS implies **larger output contractions**.
- The larger fall in prices raises the real value of nominal debt (debt-deflation), depressing demand further.

**Escaping the liquidity trap**

- **Raise the future price level** (see Chapter 9):
  - Higher expected inflation lowers the real rate, easing deleveraging and stimulating savers' consumption.
  - This channel is reinforced here because $\varphi_r$ in (10.10) is larger than in the benchmark model.
- Policies that **increase long-run output** are also expansionary (shift $AD$ to the right).

# Credit Crunch: Banking-Sector Turbulence

- Alternative post-2007/08 narrative: the recession is driven by **banking-sector stress** rather than (only) borrower deleveraging.
- Mechanism:
  - Risky assets on intermediaries' balance sheets raise funding costs.
  - Banks respond by raising equity and/or reducing leverage.
  - Capital constraints tighten $\Rightarrow$ lending supply contracts.
  - Lower credit availability depresses spending and output.
- We capture this with an explicit **lending-supply limit** in the simplified framework of the previous section.

# Modified Debt and Lending Limits

- Two agents: savers and borrowers, with preferences (10.1) and budget constraint (10.2).
- The debt constraint is augmented by an **upper bound on lending**:

$$\mathrm{L}^{high} \;\geq\; (1+r_{t-1})B^{j}_{t-1} \;\geq\; -B^{high} \;\geq\; -\frac{1}{2}\frac{\beta}{1-\beta}Y.$$

- Interpretation:
  - $\mathrm{L}^{high}$: maximum credit that lenders are willing/able to supply,
  - $B^{high}$: maximum borrowing capacity ("safe" debt limit).
- Initial steady state: $\mathrm{L}^{high} > B^{high}$, so the lending limit is slack (only the borrowing limit is relevant).

# Initial Steady State: Saving vs. Borrowing Rates

- It is useful to distinguish between the **saver's** and the **borrower's** real rate.
- Savers are unconstrained, so their steady-state real rate is pinned down by the Euler equation:

$$1 + r^s = \frac{1}{\beta^s} = \frac{1}{\beta}.$$

- Borrowers are at the borrowing limit and therefore do not satisfy their Euler equation with equality. Define an implicit (shadow) borrowing rate by evaluating (10.4) at steady-state borrower consumption:

$$1 + r^b = \frac{1}{\beta^b} > 1 + r^s.$$

- Hence there is a positive steady-state spread,

$$r^b - r^s > 0,$$

capturing that borrowers are more impatient than savers.

# Credit Crunch Shock

- Banking-sector turbulence reduces the maximum lending supply from $\mathrm{L}^{high}$ to $\mathrm{L}^{low}$.
- If $\mathrm{L}^{low} < B^{high}$, savers become constrained and lending is pinned down by $\mathrm{L}^{low}$ (the supply limit binds).
- Long-run allocations (from $t+1$ onward):

$$C^b_{t+1} = \frac{1}{2}Y - (1-\beta^b)\mathrm{L}^{low}, \qquad C^s_{t+1} = \frac{1}{2}Y + (1-\beta^b)\mathrm{L}^{low}.$$

- Since savers are constrained, the **borrowers' discount factor** is the relevant one for pricing the long-run real rate.

# Short Run (Time $t$)

- Using the flow budget constraint (10.2), with initial asset position $B^{high}$ and long-run position $\mathrm{L}^{low}$, saver consumption at time $t$ is:

$$C_t^s = \frac{1}{2}Y - \frac{\mathrm{L}^{low}}{1 + r_t^b} + B^{high}.$$

- Market clearing implies $C_t^b + C_t^s = Y$, hence borrower consumption:

$$C_t^b = \frac{1}{2}Y + \frac{\mathrm{L}^{low}}{1 + r_t^b} - B^{high}.$$

- In the short run, the market real rate is $r_t^b$ because the borrowers' Euler equation holds with equality (they are marginal in pricing).

# Borrowing Rate

- Insert the short- and long-run borrower consumption into the Euler equation (10.4) (holding with equality) to solve for the short-run borrowing rate:

$$1 + r_t^b = \frac{1}{\beta^b} \frac{\frac{1}{2}Y - \mathrm{L}^{low}}{\frac{1}{2}Y - B^{high}} > \frac{1}{\beta^b}.$$

- If $\mathrm{L}^{low} < B^{high}$, the borrowing rate **jumps up**.
    - A negative lending-supply shock therefore raises the effective rate faced by borrowers.

# Shadow Saving Rate

- With a binding lending limit, savers cannot freely increase their asset holdings. Hence the savers' Euler condition (10.4) holds with a **strict reverse inequality**:

$$U_c(C_t^s) \; > \; \beta^s(1 + r_t^b)\, U_c(C_{t+1}^s).$$

- Define the **shadow saving rate** $r_t^s$ as the interest rate that would make the savers' Euler equation hold with equality:

$$U_c(C_t^s) = \beta^s(1 + r_t^s)\, U_c(C_{t+1}^s).$$

- In the model, $r_t^s$ satisfies:

$$1 + r_t^s = \frac{1}{\beta^s} \frac{\frac{1}{2}Y + \mathrm{L}^{low}}{\frac{1}{2}Y + B^{high}} + \frac{1}{\frac{1}{2}Y + B^{high}} \left( \frac{1 + r_t^s}{1 + r_t^b} - \frac{\beta^b}{\beta^s} \right) \mathrm{L}^{low}. \tag{10.14}$$

- The RHS is increasing in $(1 + r_t^s)$ and lies below the 45° line at $(1 + r_t^s) = 1/\beta^s$, implying the fixed point:

$$1 + r_t^s < \frac{1}{\beta^s}.$$

# Intuition: Borrowing vs. Saving Rates

- A contraction in credit supply raises the borrowing rate above its steady state:
$$1 + r_t^b > \frac{1}{\beta^b}.$$
- At the same time, the spread widens: $r_t^b - r_t^s$ increases when lending falls.
- In (10.14), the second term becomes negative, which pushes the shadow saving rate down:
$$1 + r_t^s \downarrow .$$

**Interpretation:**

- Savers are forced to lend less, so they must be induced to **save less and consume more** $\Rightarrow$ lower $r_t^s$.
- Borrowers must **borrow less and consume less** to clear goods markets $\Rightarrow$ higher $r_t^b$.
- This mirrors the deleveraging case: borrowing rate increases while the saving rate falls.

# Deleveraging and Credit Crunch: Isomorphism

- Deleveraging shocks and credit-crunch shocks imply the same qualitative movements:
  - $r_t^b$ rises to discourage borrowing and compress borrower spending,
  - $r_t^s$ falls to encourage saver consumption,
  - the spread $r_t^b - r_t^s$ widens.
- A sufficiently large contraction can push the saving real rate below zero, potentially generating a liquidity trap.

# Income Variation and the Interest Rate Spread

- Credit spreads can also move because of income changes (not only borrowing/lending limits).
- **Temporary aggregate income increase** affecting both agents equally:
  - Constrained borrowers consume the additional income.
  - Savers would like to save more $\Rightarrow$ the saving rate must fall to stimulate their consumption.
  - The shadow borrowing rate falls proportionally, so the spread is (approximately) unchanged.
- **Unequal income changes / transfers** can change the spread:
  - E.g. borrowers' income rises while savers' income falls.
  - Borrowers consume the transfer, savers would like to borrow (reduce assets).
  - Goods-market clearing requires the saving rate to rise (discourage savers from borrowing).
  - The borrowing rate falls, so the spread decreases.

# Two Euler Equations with Default Risk

- Consider again a two-agent economy with preferences given by (10.5).
- Savers and borrowers are **unconstrained**, so their Euler equations hold with equality.

- Saver Euler equation:

$$\exp(-zC_t^s) = \beta^s(1 + i_t^s)\mathbb{E}_t\left\{\frac{P_t}{P_{t+1}}\exp(-zC_{t+1}^s)\right\}. \tag{10.15}$$

- Borrower Euler equation with default risk:

$$\exp(-zC_t^b) = \beta^b(1 + i_t^b)\mathbb{E}_t\left\{\frac{P_t}{P_{t+1}}(1 - \omega_{t+1})\exp(-zC_{t+1}^b)\right\}, \tag{10.16}$$

where $i_t^s$ and $i_t^b$ are nominal interest rates faced by savers and borrowers, and $\omega_{t+1} \in [\omega_{\min}, \omega_{\max}]$ is the default rate with $0 < \omega_{\min} < \omega_{\max} < 1$.

# Log-Linearization and Aggregation

- A log-linear approximation of (10.15)–(10.16) yields:

$$C_t^s = \mathbb{E}_t C_{t+1}^s - \frac{1}{z}\left(\hat{\imath}_t^s - \mathbb{E}_t(\pi_{t+1} - \pi)\right), \qquad C_t^b = \mathbb{E}_t C_{t+1}^b - \frac{1}{z}\left(\hat{\imath}_t^b - \mathbb{E}_t\hat{\omega}_{t+1} - \mathbb{E}_t(\pi_{t+1} - \pi)\right),$$

  where hats denote log-deviations and $\hat{\omega}_{t+1}$ is the deviation of the default rate from steady state.

- Using aggregation $Y_t = \chi C_t^s + (1-\chi)C_t^b$, we obtain:

$$\hat{Y}_t = \mathbb{E}_t \hat{Y}_{t+1} - \sigma\left(\hat{\imath}_t^s - \mathbb{E}_t(\pi_{t+1} - \pi) + (1-\chi)\left(\hat{\imath}_t^b - \mathbb{E}_t\hat{\omega}_{t+1} - \hat{\imath}_t^s\right)\right), \tag{10.17}$$

  where $\sigma \equiv \frac{1}{zY}$.

# Interest-Rate Spread as a Preference Shock

- Equation (10.17) is isomorphic to the Chapter 5 AD equation (with $G_t = 0$).
- Identify the policy-relevant nominal rate with the saver rate:

$$\hat{\imath}_t \equiv \hat{\imath}_t^s .$$

- The remaining wedge behaves like a preference (demand) shock:

$$E_t \Delta \hat{\xi}_{t+1} \equiv E_t(\hat{\xi}_{t+1} - \hat{\xi}_t) = (1-\chi)\left(\hat{\imath}_t^b - \mathbb{E}_t \hat{\omega}_{t+1} - \hat{\imath}_t^s\right).$$

- Hence a higher **default-adjusted credit spread** lowers aggregate demand and output, exactly like a negative preference shock.

# Natural (Efficient) Real Rate with Spreads

- Rewrite AD in efficient-gap form (same $\hat{Y}_t^e$ as Chapter 7):

$$\hat{Y}_t - \hat{Y}_t^e = \mathbb{E}_t(\hat{Y}_{t+1} - \hat{Y}_{t+1}^e) - \sigma\left(\hat{\imath}_t^s - \mathbb{E}_t(\pi_{t+1} - \pi) - r_t^e\right). \tag{10.18}$$

- The frictionless (efficient) real rate is:

$$r_t^e \equiv \mathbb{E}_t\left\{\frac{1+\eta}{1+\sigma\eta}\big(\hat{A}_{t+1} - \hat{A}_t\big) - (1-\chi)\big(\hat{\imath}_t^b - \mathbb{E}_t\hat{\omega}_{t+1} - \hat{\imath}_t^s\big)\right\}. \tag{10.19}$$

- A rise in the default-adjusted credit spread lowers $r_t^e$.
    - If the spread is endogenous, monetary policy can influence $r_t^e$ indirectly through its effects on credit conditions.

# AS Equation

- With exponential utility and Cobb–Douglas aggregation, the AS block takes the same reduced form as in Chapter 5/7:

$$\pi_t - \pi = \kappa(\hat{Y}_t - \hat{Y}_t^e) + v_t + \beta \mathbb{E}_t(\pi_{t+1} - \pi). \tag{10.20}$$

- The parameters $\kappa$ and $v_t$ are as defined earlier.
- **Key message:** Heterogeneity affects the **IS/AD side** through spreads, while the Phillips-curve structure remains standard.

# Financial Intermediaries

- A stylized intermediary (lives $t$ and $t+1$) lends $Z_t$, funded with deposits $D_t$ and equity $N_t$. Its balance-sheet constraint is:

$$Z_t = D_t + N_t\big(1 - \varsigma(Z_t)\big), \tag{10.21}$$

where $\varsigma(Z_t)$ is an extra cost per unit of equity, non-decreasing in $Z_t$.

- At $t+1$, nominal profits are:

$$\Psi^I_{t+1} = (1 + i^b_t)(1 - \omega_{t+1})Z_t - (1 + i^s_t)D_t. \tag{10.22}$$

- Lending earns $i^b_t$ but faces random default $\omega_{t+1}$,
- Deposits pay the saving rate $i^s_t$.

# Limited Liability and Equity Lower Bound

- Limited liability requires $\Psi^I_{t+1} \geq 0$ in all states; in the worst case $\omega_{\max}$:
$$(1+i^b_t)(1-\omega_{\max})Z_t \geq (1+i^s_t)D_t. \tag{10.23}$$

- Using (10.21) to substitute for $D_t$ gives:
$$\Big[(1+i^b_t)(1-\omega_{\max}) - (1+i^s_t)\Big]Z_t \geq -(1+i^s_t)\,N_t\big(1-\varsigma(Z_t)\big).$$

- Equivalently, equity must satisfy the lower bound:
$$N_t \geq \frac{1}{1-\varsigma(Z_t)}\left[1 - \frac{1+i^b_t}{1+i^s_t}(1-\omega_{\max})\right]Z_t. \tag{10.24}$$

- Intuition: intermediaries must hold enough equity to absorb losses under worst-case default.

# Intermediary Rents

- Intermediaries are owned by savers and maximize rents:

$$\mathcal{R}_t = E_t\left\{\tilde{R}^s_{t,t+1}\,\Psi^I_{t+1}\right\} - N_t,$$

where the saver SDF is

$$\tilde{R}^s_{t,t+1} = \beta^s \frac{\exp(-zC^s_{t+1})}{\exp(-zC^s_t)}\,\frac{P_t}{P_{t+1}}.$$

- Using (10.22) and the saver Euler equation (10.15), rents can be written as:

$$\mathcal{R}_t = \frac{1+i^b_t}{1+i^s_t}\left(1-\tilde{E}_t\omega_{t+1}\right)Z_t - D_t - N_t, \tag{10.25}$$

where expected default is measured under the “risk-neutral” probability:

$$\tilde{E}_t\omega_{t+1} \equiv (1+i^s_t)\,E_t\left\{\tilde{R}^s_{t,t+1}\,\omega_{t+1}\right\}. \tag{10.26}$$

# Zero Rents and the Interest-Rate Spread

- Substitute $D_t$ from the balance sheet (10.21) into rents (10.25):

$$\mathcal{R}_t = \left( \frac{1 + i_t^b}{1 + i_t^s} \left(1 - \tilde{E}_t \omega_{t+1}\right) - 1 \right) Z_t - \varsigma(Z_t)\, N_t .$$

- Impose limited liability (10.24) with equality to eliminate $N_t$, obtaining:

$$\mathcal{R}_t = Z_t \left[ \left( \frac{1 + i_t^b}{1 + i_t^s} \left(1 - \tilde{E}_t \omega_{t+1}\right) - 1 \right) - \frac{\varsigma(Z_t)}{1 - \varsigma(Z_t)} \left( 1 - \frac{1 + i_t^b}{1 + i_t^s} (1 - \omega_{\max}) \right) \right]. \tag{10.27}$$

- Competition implies $\mathcal{R}_t = 0$, pinning down the spread:

$$\frac{1 + i_t^b}{1 + i_t^s} = \frac{1}{(1 - \varsigma(Z_t)) \left(1 - \tilde{E}_t \omega_{t+1}\right) + \varsigma(Z_t)\,(1 - \omega_{\max})}. \tag{10.28}$$

# Equity-to-Loans Ratio

- Using (10.24) and the spread condition (10.28), the equity-to-loans ratio is:

$$\frac{N_t}{Z_t} = \frac{1 + i_t^b}{1 + i_t^s} \left( \omega_{\max} - \tilde{E}_t \omega_{t+1} \right).$$

**Interpretation**

- $\frac{N_t}{Z_t}$ increases with the spread and with the gap between worst-case and expected loan losses.
- Intermediaries must hold equity to cover losses under the worst-case default scenario.

**Special case:**

- If $\varsigma(\cdot) = 0$, then (10.28) depends only on expected default risk. In a log-linear approximation, this is not a (first-order) source of perturbation in the AD equation (10.17).

# Spread Function

- The spread can be summarized as a reduced-form function:

$$\frac{1+i_t^b}{1+i_t^s} = s\left(Z_t, \omega_{\max}, \tilde{E}_t \omega_{t+1}\right).$$

- Comparative statics:
  - $s(\cdot)$ is non-decreasing in $Z_t$ if $\varsigma(Z_t)$ is non-decreasing (higher intermediation increases equity-raising costs).
  - Higher $\varsigma(Z_t)$ increases the spread.
  - Higher worst-case loss $\omega_{\max}$ or higher expected loss $\tilde{E}_t \omega_{t+1}$ increases the spread.

# Model Implications and Extensions

- The stylized model is consistent with broader mechanisms:
  - Higher intermediation volumes require a higher spread to compensate for risk and equity costs.
  - The worst-case default rate $\omega_{\max}$ can be endogenized (e.g. as a function of borrower income or collateral values).
  - In good macro states, a lower $\omega_{\max}$ reduces spreads.
- These channels highlight the interaction between spreads, policy, and macro outcomes, motivating richer models of spread determination.
- Examples where spreads depend on aggregate debt and interact with monetary policy: Benigno, Eggertsson, and Romei (2020) and Curdia and Woodford (2010) and Curdia and Woodford (2011).

# Why Policy Must Watch Credit Spreads

- From (10.18)–(10.19):
  - Stabilization requires moving the policy-relevant rate $i_t^s$ in line with the efficient real rate $r_t^e$.
  - If the default-adjusted spread rises, $r_t^e$ **falls** even absent preference shocks.
  - Therefore the policy rate must fall by **more**; at the ZLB this may be impossible.
- **Conclusion:** Credit spreads provide real-time information about the "right" real rate and the stance of policy.

# Forward Guidance with Endogenous Spreads

- Chapter 9: Forward guidance (guidance about future short rates) helps escape a liquidity trap.
- With endogenous spreads, the exit from ZLB policies can occur earlier than when the disturbance is purely exogenous:
  - Policies that improve borrowers' repayment capacity can shorten deleveraging.
  - Policies that reduce default probabilities can dampen the initial banking shock and its propagation.
- Endogenous spreads can also attenuate the sensitivity of output to movements in $\hat{\imath}_t^s$.

# Optimal Policy

- A second-order approximation of welfare around the efficient steady state yields the quadratic loss:

$$L_{t_0}^{CB} = \frac{1}{2}\mathbb{E}_{t_0}\left\{\sum_{t=t_0}^{\infty}\beta^{t-t_0}\left[(\hat{Y}_t - \hat{Y}_t^e)^2 + \chi(1-\chi)\Lambda_c(\hat{C}_t^b - \hat{C}_t^s)^2 + \Lambda_\pi(\pi_t - \pi)^2\right]\right\}. \quad (10.29)$$

- Relative to Chapter 7, policy now also cares about **risk sharing**: deviations $(\hat{C}_t^b - \hat{C}_t^s)$ capture distributional distortions across borrowers and savers.
- Even with exogenous spreads, policy should account for financial conditions (Curdia and Woodford 2016).
- With deleveraging or banking turbulence, distributional concerns tilt optimal policy in a more expansionary direction (supporting borrowers' consumption).

# Unconventional Monetary Policy

- The framework supports **credit-easing** policies (Chapter 9): they can compress credit spreads and mitigate the decline in the efficient real rate during crises.
- Acting as lender of last resort, the central bank can counteract spread widening, dampening the impact of financial distress on inflation and output.
- Credit easing can help achieve output and inflation objectives and reduce both the **severity** and the **duration** of a liquidity trap.

# Main Takeaways

- **Liquidity traps can be endogenous:** Deleveraging and credit crunches lower the natural real rate and can push it below zero.
- **Deleveraging:** Borrowers cut spending; full-employment demand requires savers to spend more, implying a lower natural rate.
- **Credit crunch:** Banking stress tightens lending and widens credit spreads, depressing demand and lowering the efficient real rate.
- With **sticky prices** and the **ZLB**, the policy rate may not fall enough, generating a Keynesian slump and debt-deflation dynamics.
- **Policy lesson:** Monitor spreads and use forward guidance / credit easing to compress spreads and shorten liquidity traps.

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